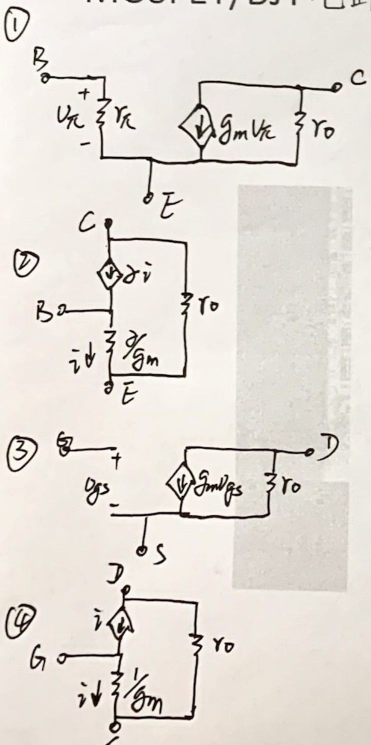
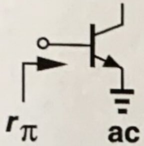
Input and Output Impedances

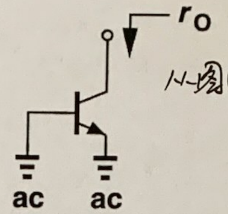
验证常见结构的小信号输入输出电阻表达式（用小信号模型替代MOSFET/BJT，再求输入输出电阻，求输出电阻时注意信号源短路。熟记该图有利于快速直观地分析MOSFET/BJT电路）

作业

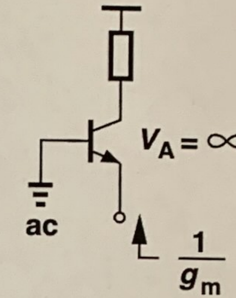




从图①直接得出



从图①直接得出



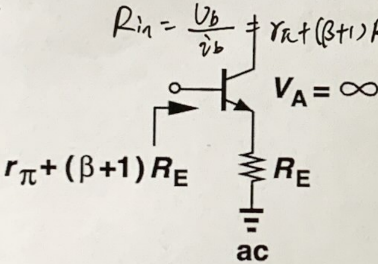
根据图②，不考虑 r_o，α近似为1，可得到

图①

$v_b = i_b\cdot r_\pi + i_e\cdot R_E$

$= i_b[r_\pi+(\beta+1)R_E]$

$R_{in} = \frac{v_b}{i_b} = r_\pi+(\beta+1)R_E$



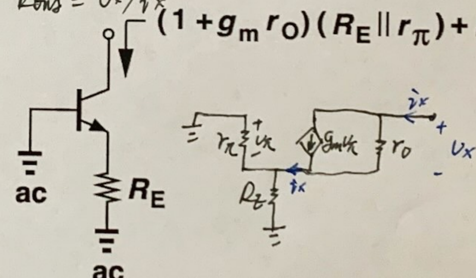
$-v_\pi = i_x(r_\pi//R_E)$

$v_x = -v_\pi + (i_x - g_m v_\pi)\cdot r_o$

$= i_x(r_\pi//R_E) + i_x\cdot r_o + i_x\cdot g_m r_o(r_\pi//R_E)$

$R_{out} = v_x/i_x$

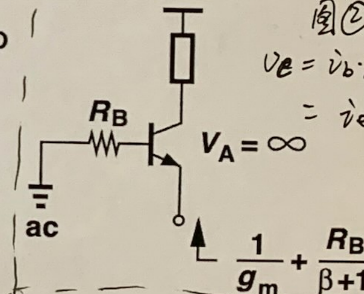
$(1+g_m r_O)(R_E \| r_\pi)+r_O$



图②

$v_e = i_b\cdot R_B + \frac{1}{g_m}\cdot i_e$

$= i_e\left(\frac{R_B}{\beta+1}+\frac{1}{g_m}\right)$

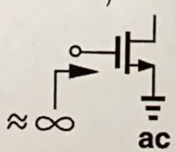


$v_x = i_x\cdot R_S + (i_x - i)\cdot r_o$

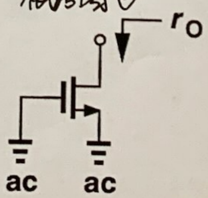
$-i = \frac{i_x\cdot R_S}{1/g_m}$

$\Rightarrow \frac{v_x}{i_x} = R_S + r_o + g_m R_S r_o$

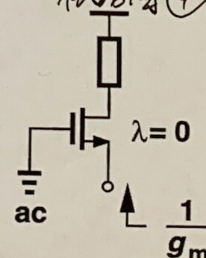
栅极开路



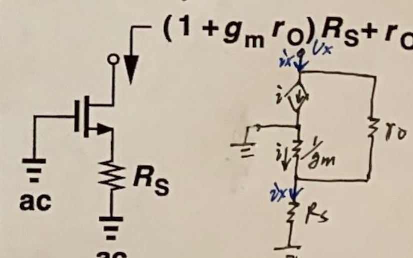
根据图③



根据图④



$(1+g_m r_O)R_S+r_O$



7.4 For the amplifier in Fig. 7.10, let $V_{DD} = 5$ V, $R_D = 10$ kΩ, $V_t = 1$ V, $k_n' = 20\ \mu\text{A/V}^2$, $W/L = 20$, $V_{GS} = 2$ V, and $\lambda = 0$.

(a) Find the dc current I_D and the dc voltage V_{DS}. $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{ov}^2 = 200\ \mu A$ $V_{DS} = 5 - 10k \cdot 0.2m = 3V$

(b) Find g_m. $g_m = \frac{I_D}{V_{ov}/2} = \frac{0.2mA}{0.5V} = 0.4\ mA/V$

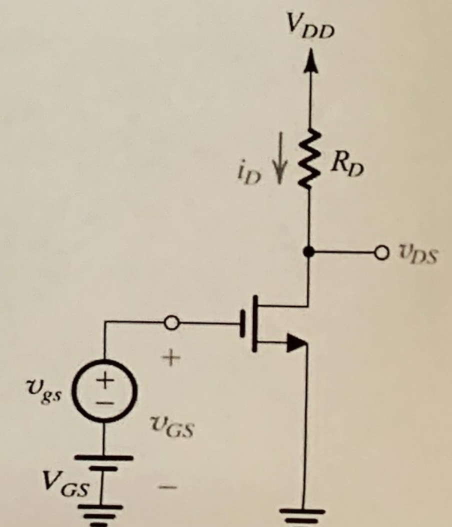
(c) Find the voltage gain. $A_v = -g_m R_D = -4\ V/V$

(d) If $v_{gs} = 0.2 \sin \omega t$ volts, find v_{ds} assuming that the small-signal approximation holds. What are the minimum and maximum values of v_{DS}? $V_{ds} = V_{gs} \cdot A_v = -0.8 \sin\omega t\ V$, $V_{DS} + 0.8 = 3.8V$ $V_{DS} - 0.8V = 2.2V$

(e) Use Eq. (7.28) to determine the various components of i_D. Using the identity $\sin^2 \omega t = \frac{1}{2} - \frac{1}{2}\cos 2\,\omega t$, show that there is a slight shift in I_D (by how much?) and that there is a second-harmonic component (i.e., a component with frequency $2\,\omega$). Express the amplitude of the second-harmonic component as a percentage of the amplitude of the fundamental. (This value is known as the second-harmonic distortion.)

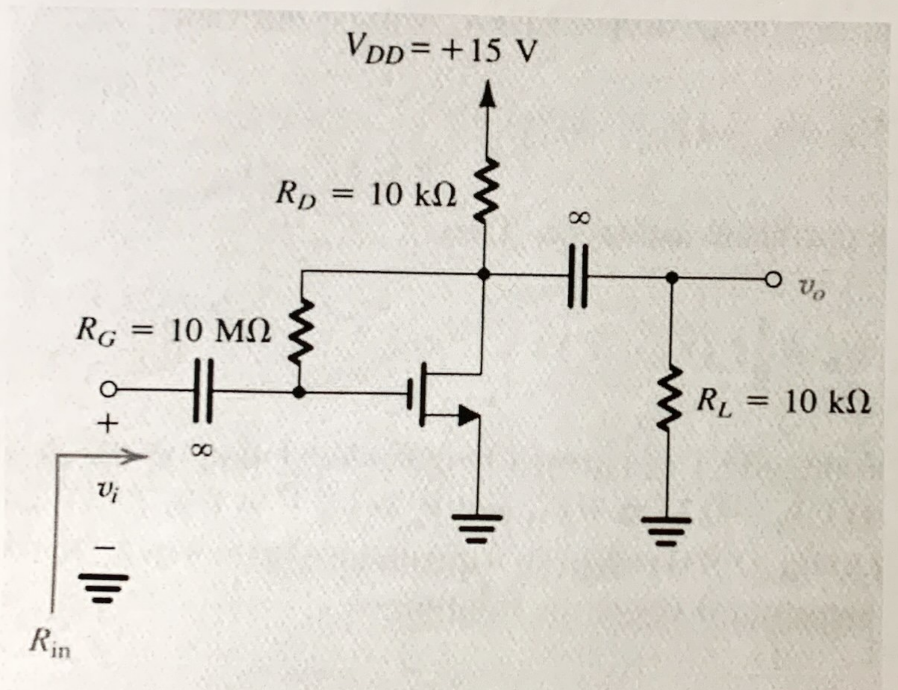
Ans. (a) 0.2 mA, 3 V; (b) 0.4 mA/V; (c) –4 V/V; (d) $v_{ds} = -0.8 \sin \omega t$ volts, 2.2 V, 3.8 V; (e) $i_D = (204 + 80 \sin \omega t - 4 \cos 2\,\omega t)\ \mu$A, 5%

$$i_D = \frac{1}{2}k_n(V_{GS} + v_{gs} - V_t)^2$$
$$= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \qquad (7.28)$$



D7.10 Consider the amplifier circuit of Fig. 7.16(a) without the load resistance R_L and with channel-length modulation neglected. Let $V_{DD} = 5$ V, $V_t = 0.7$ V, and $k_n = 1$ mA/V^2. Find V_{OV}, I_D, R_D, and R_G to obtain a voltage gain of -25 V/V and an input resistance of 0.5 M Ω. What is the maximum allowable input signal, $\hat{v}_i$?

Ans. 0.319 V; 50.9 μA; 78.5 kΩ; 13 MΩ; 27 mV



(a)

$$\left.\begin{aligned} R_{in} &= \frac{R_G}{1+g_m R_D} = 0.5M \\ g_m R_D &= 25 \end{aligned}\right\} \Rightarrow R_G = 13M\Omega$$

$$\left.\begin{aligned} I_D &= \tfrac{1}{2} k_n V_{ov}^2 \\ k_n V_{ov} &= \frac{25}{R_D} \\ V_{ov} &= 5 - R_D I_D - 0.7 \end{aligned}\right\} \Rightarrow V_{ov} = 0.319\,V$$

$\therefore I_D = 50.9\,\mu A$

$R_D = 78.5\,k\Omega$

$$V_{i,max} = \frac{V_t}{|A|+1} = \frac{0.7}{26} = 27mV$$

7.20 The transistor in Fig. E7.20 is biased with a constant current source $I = 1$ mA and has $\beta = 100$ and $V_A = 100$ V.

(a) Neglecting the Early effect, find the dc voltages at the base, emitter, and collector.

(b) Find g_m, r_π, and r_o.

(c) If terminal Z is connected to ground, X to a signal source v_{sig} with a source resistance $R_{sig} =$ 2 kΩ, and Y to an 8-kΩ load resistance, use the hybrid-π model shown in Fig. 7.26 to draw the small-signal equivalent circuit of the amplifier. (Note that the current source I should be replaced with an open circuit.) Calculate the overall voltage gain v_y/v_{sig}. If r_o is neglected, what is the error in estimating the gain magnitude? (*Note:* An infinite capacitance is used to indicate that the capacitance is large enough to act as a short circuit at all signal frequencies of interest. However, the capacitor still blocks dc.)

$I_B = \frac{1mA}{100} = 10\mu A$

$V_B = 0 - 10K \cdot 10\mu = -0.1V$

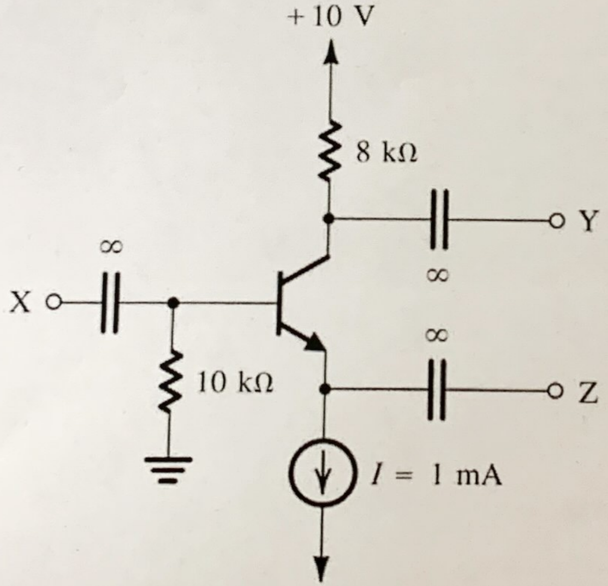
$V_E = V_B - 0.7 = -0.8V$

$V_C = 10 - 8K \cdot 1m = 2V$

$g_m = \frac{I_C}{V_T} = \frac{1}{25}$ S

$r_\pi = \frac{\beta}{g_m} = 2.5k\Omega$

$r_o = \frac{|V_A|}{I_C} = 100k\Omega$



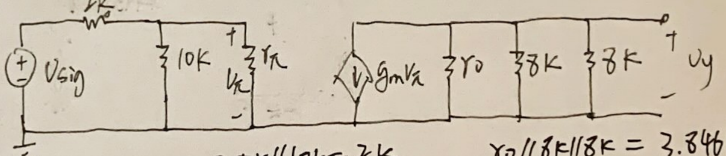


Figure E7.20

$2.5K // 10K = 2K$

$\frac{V_\pi}{V_{sig}} = \frac{1}{2}$

$r_o // 8K // 8K = 3.846K$

与4K的误差为4%

$\frac{V_y}{V_{sig}} = \frac{1}{2} \cdot g_m (r_o // 8K // 8K)$

$= 76.92$ V/V

Ans. (a) –0.1 V, –0.8 V, +2.1 V; (b) 40 mA/V, 2.5 kΩ, 100 kΩ; (c) –77 V/V, +3.9%